IRC Results — Full Sella_hip Sweep

sella_hip IRC on gad_dt003 converged TSs, 2026-04-16

Scope

Full IRC validation of the best GAD variant (`gad_dt003`, projected GAD with fixed $\mathrm{d}t = 0.003$) against the `Transition1x` train split reactant/product labels using `sella_hip` — Sella's IRC algorithm with the BFGS Hessian update replaced by HIP's analytic mass-weighted, Eckart-projected Hessian after every inner step.

Pipeline.

1. Start from a `Transition1x` labeled TS, perturb by isotropic Gaussian noise (6 noise levels: 10/30/50/100/150/200
2. Run `gad_dt003`; at the first step where Eckart-projected $n_{\mathrm{neg}} = 1$ and $\|\mathbf{F}\| < 0.01\,\mathrm{eV/Å}$, take the trajectory coordinates at that step. No refinement, no re-gating.
3. From that TS, run `sella_hip` IRC forward/reverse for up to 500 steps.
4. Compare forward/reverse endpoints to labeled reactant/product using element-aware Kabsch+Hungarian RMSD and bond-graph isomorphism.
5. Re-evaluate HIP's vibrational Hessian at both endpoints to confirm each is a true minimum ($n_{\mathrm{neg,vib}} = 0$).

Primary metric. *TOPO-intended*: both directions match (reactant, product) by bond-graph isomorphism. RMSD-intended uses the stricter $< 0.3\,\text{Å}$ threshold on the same direction-pair. Both labels are direction-agnostic (forward/reverse can swap).

Headline Numbers

1462 IRC runs across 6 noise levels, zero errors, average 11.9 s wall per sample (forward+reverse IRC).

Noise	n	TOPO-int%	TOPO-half%	RMSD-int%	RMSD-half%	wall_{avg} (s)
10 pm	284	94.4	5.6	64.4	32.0	12.6
30 pm	283	94.3	5.6	64.3	32.5	12.4
50 pm	276	93.1	6.5	64.9	31.2	12.3
100 pm	262	93.9	5.7	65.6	30.5	12.2
150 pm	214	90.7	9.3	66.8	29.9	11.8
200 pm	143	80.4	18.2	60.8	34.3	10.4
all	1462	**92.1**	7.9	**64.7**	31.8	11.9

Notation for the three criteria. Let $\mathbf{x}_f, \mathbf{x}_r$ be the IRC forward/reverse endpoint coordinates; $\mathbf{x}_R, \mathbf{x}_P$ the `Transition1x`-labeled reactant/product. All are $3N$-vectors for the same atom ordering/symbols.

- $d^{a\to b}$ = element-labeled Kabsch+Hungarian RMSD between $\mathbf{x}_a$ and $\mathbf{x}_b$. Computed for all four pairs: $(f,R),(f,P),(r,R),(r,P)$.
- $g(\mathbf{x})$ = element-labeled bond graph from covalent-radius heuristic. $g_a \cong g_b$ is graph isomorphism that preserves atom labels. Computed for all four pairs.
- $n_{\text{neg}}(\mathbf{x})$ = number of negative eigenvalues of HIP's analytic Hessian *after* mass-weighting and projecting out the 6 translation/rotation Eckart modes. $n_{\text{neg}} = 0$ iff $\mathbf{x}$ is at a true vibrational minimum.

Criterion 1 — bond-graph topology.

$$\begin{aligned}
\texttt{TOPO-intended} &\equiv (g_f \cong g_R \ \wedge\ g_r \cong g_P) \ \vee\ (g_f \cong g_P \ \wedge\ g_r \cong g_R) \\
\texttt{TOPO-half} &\equiv \big(g_f \cong g_R \vee g_f \cong g_P \vee g_r \cong g_R \vee g_r \cong g_P\big) \ \wedge\ \neg\texttt{TOPO-intended} \\
\texttt{TOPO-unintended} &\equiv \neg\texttt{TOPO-intended} \ \wedge\ \neg\texttt{TOPO-half}
\end{aligned}$$

Direction-agnostic: the forward endpoint may land on either R or P. A single missing match anywhere is a "half"; no matches at all is "unintended".

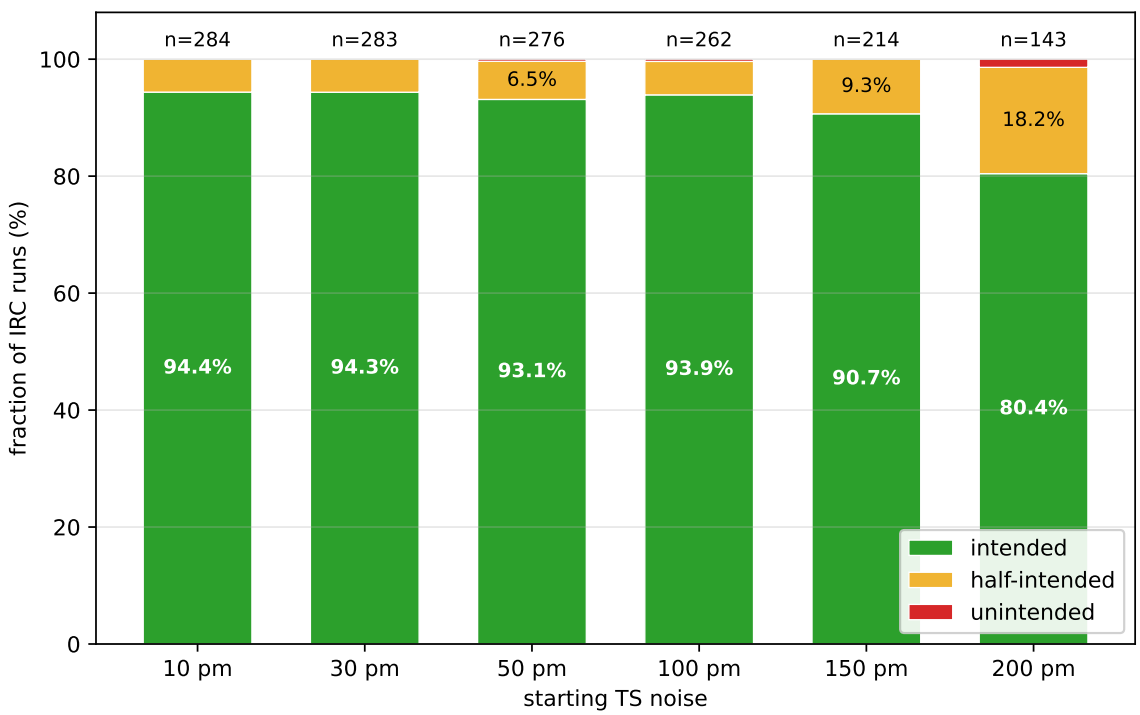

Figure 1: `sella_hip` IRC on `gad_dt003` converged TSs — bond-graph topology criterion. Denominator = `gad_dt003` converged TSs at each noise level; intended = forward+reverse endpoints match the labeled (R, P) pair by element-labeled bond-graph isomorphism. Flat at ∼94% through 100 pm; steps down to 90.7% at 150 pm; drops to 80.4% at 200 pm.

Criterion 2 — strict RMSD (< 0.3 Å).

$$d_{\to R} = \min(d^{f\to R},\ d^{r\to R}) \qquad d_{\to P} = \min(d^{f\to P},\ d^{r\to P})$$

$$\begin{aligned}
\texttt{RMSD-intended} &\equiv (d_{\to R} < 0.3\ \text{Å}) \ \wedge\ (d_{\to P} < 0.3\ \text{Å}) \\
\texttt{RMSD-half} &\equiv \big((d_{\to R} < 0.3) \vee (d_{\to P} < 0.3)\big) \ \wedge\ \neg\texttt{RMSD-intended} \\
\texttt{RMSD-unintended} &\equiv \neg\texttt{RMSD-intended} \ \wedge\ \neg\texttt{RMSD-half}
\end{aligned}$$

Also direction-agnostic (we take the min over forward/reverse for each target). Note: RMSD-intended is strictly stronger than TOPO-intended — any geometry close enough to match RMSD will also match bond graph.

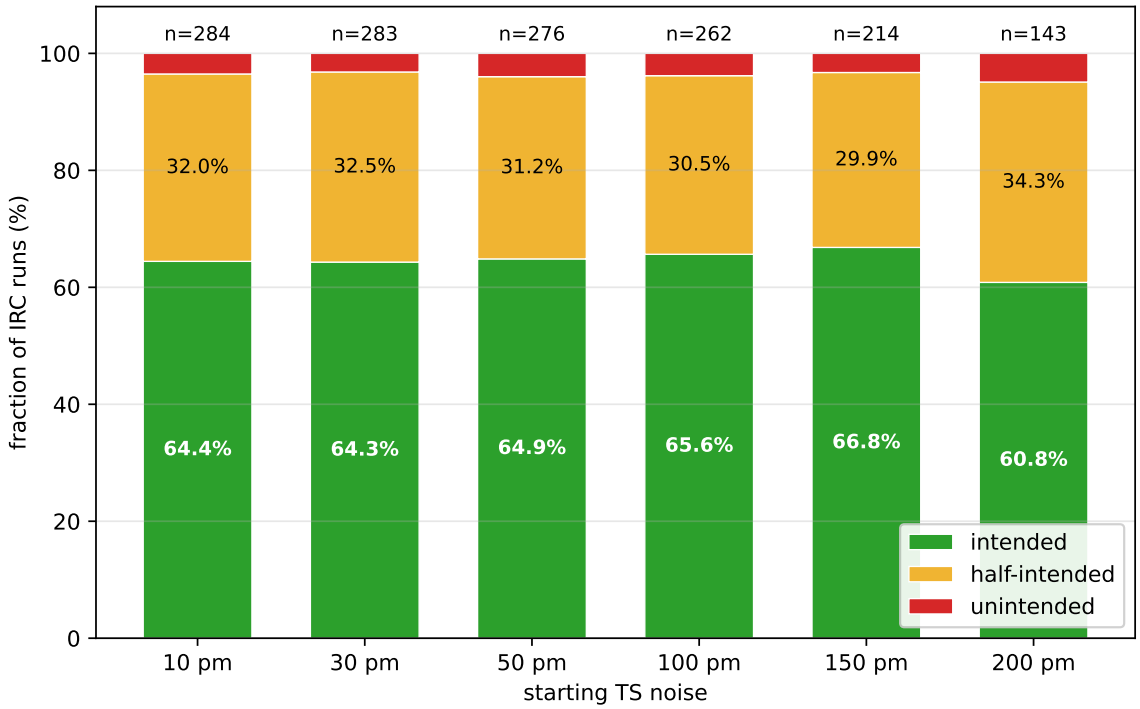

Figure 2: `sella_hip` IRC on `gad_dt003` converged TSs — strict RMSD criterion (< 0.3 Å). Same denominator as above; intended = both endpoints match the labeled (R, P) pair by Kabsch+Hungarian RMSD below 0.3 Å. Roughly flat at 60–67% across noise levels; the gap to topology is absorbed mostly by the half-intended tier.

Criterion 3 — endpoint vibrational quality.

$$\begin{aligned}
\texttt{both_at_minimum} &\equiv \big(n_{\text{neg}}(\mathbf{x}_f) = 0\big) \;\wedge\; \big(n_{\text{neg}}(\mathbf{x}_r) = 0\big) \\
\texttt{only_one_at_minimum} &\equiv \big(n_{\text{neg}}(\mathbf{x}_f) = 0\big) \;\oplus\; \big(n_{\text{neg}}(\mathbf{x}_r) = 0\big) \\
\texttt{neither_at_minimum} &\equiv \big(n_{\text{neg}}(\mathbf{x}_f) > 0\big) \;\wedge\; \big(n_{\text{neg}}(\mathbf{x}_r) > 0\big)
\end{aligned}$$

This does not reference the labeled reactant/product — it asks only whether the IRC integrator managed to descend to a real minimum on HIP's PES in each direction. "neither at minimum" is a *ridge-stall*: the integrator halted while one or both endpoints still had a non-trivial unstable mode.

Endpoint Vibrational Quality

For each IRC endpoint we re-compute HIP's analytic Hessian, project out translation/rotation (Eckart), and inspect the vibrational spectrum. An endpoint is a *true minimum* iff $n_{\text{neg,vib}} = 0$ *and* $\min \lambda_{\text{vib}} > 0$.

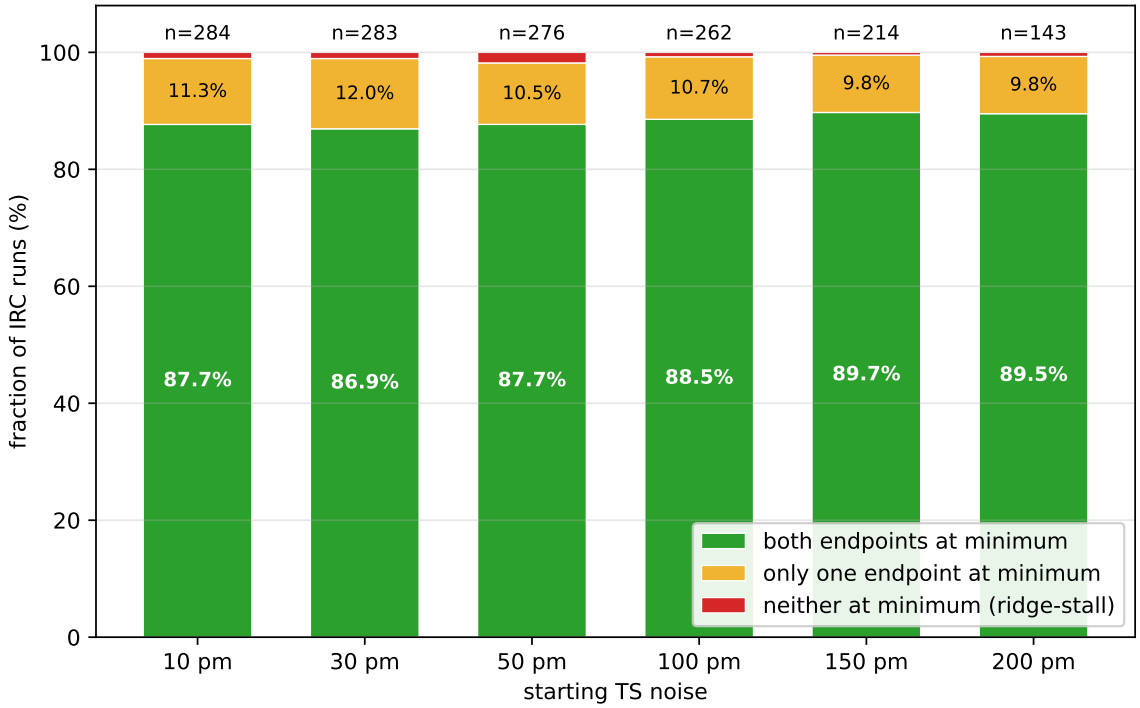

Figure 3: `sella_hip` IRC on `gad_dt003` converged TSs — endpoint vibrational quality. Same denominator; bars show the number of IRC endpoints at a true vibrational minimum ($n_{\mathrm{neg,vib}} = 0$ on the Eckart-projected Hessian at the endpoint). Both-at-minimum rises gently from 87.7% at 10 pm to 89.5% at 200 pm. Ridge-stalls are 0.5–1.8% — a tight upper bound on pure integrator failure.

Noise	fwd at min%	rev at min%	both at min%	neither at min%	any-dir match†%
10 pm	91.5	95.1	87.7	1.1	100.0
30 pm	91.2	94.7	86.9	1.1	100.0
50 pm	92.0	93.8	87.7	1.8	99.6
100 pm	93.1	94.7	88.6	0.8	99.6
150 pm	93.5	95.8	89.7	0.5	100.0
200 pm	93.0	95.8	89.5	0.7	98.6

†forward or reverse matched *some* label by bond graph.

Topology vs RMSD Gap

Bond-graph topology is far more lenient than 0.3 Å RMSD. Among the 1347 TOPO-intended runs:

Noise	TOPO-int	also RMSD-int	only TOPO	median RMSD →R (Å)
10 pm	268	183	85	0.009
30 pm	267	182	85	0.009
50 pm	257	179	78	0.009
100 pm	246	172	74	0.009
150 pm	194	143	51	0.008
200 pm	115	87	28	0.008

The median RMSD to the labeled reactant *among TOPO-intended runs* is 0.008–0.009 Å. The distribution is clearly bimodal: most topology-intended runs land exactly on the label, and a smaller cluster reaches the same bond-graph via a different conformer.

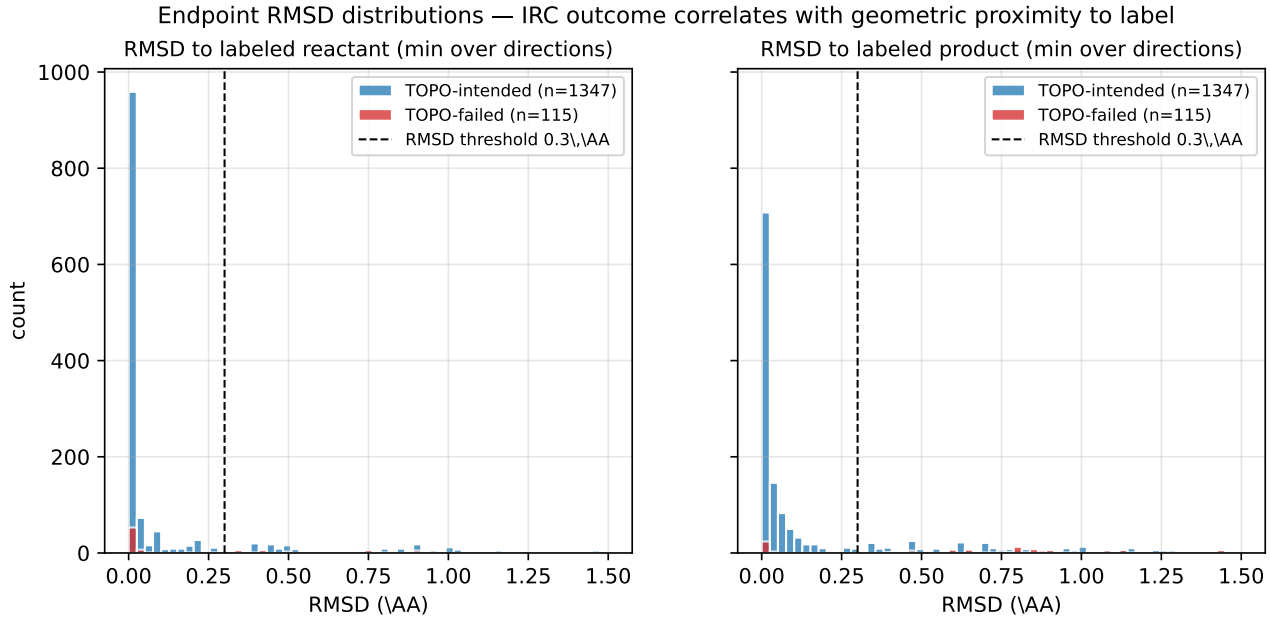

Figure 4: Endpoint RMSD to labeled reactant (left) and product (right), split by IRC outcome. TOPO-intended runs are tightly clustered near zero. TOPO-failed runs show a broad tail at 1–2 Å — consistent with IRC descending to a different minimum on the PES.

Directional Symmetry

Noise	fwd match%	rev match%	only-fwd%	only-rev%	both%	neither%
10 pm	96.8	97.5	2.5	3.2	94.4	0.0
30 pm	97.2	97.2	2.8	2.8	94.4	0.0
50 pm	96.7	96.0	3.6	2.9	93.1	0.4
100 pm	97.0	96.6	3.1	2.7	93.9	0.4
150 pm	98.1	94.9	5.1	1.9	93.0	0.0
200 pm	93.7	90.9	7.7	4.9	86.0	1.4

The forward and reverse directions hit labeled endpoints at nearly identical rates (difference $\leq$3 pp at every noise level except 150 pm, where forward is slightly more reliable). "Only one direction matched" is always $\leq$8%. This is the expected symmetry for a well-behaved IRC integrator; the HIP-injected Sella shows no directional bias.

Systematic Failures — Same Samples Fail at Every Noise Level

Fifteen samples are present at $\geq$4 noise levels and fail TOPO at every one of them. Ten of them fail at $\geq$5 of the 6 levels:

sid	formula	fails/total	median best RMSD (Å)	median $[n_{\text{neg,fwd}}, n_{\text{neg,rev}}]$	class
48	C2H3N3	6/6	0.00	[0, 0]	valid-but-wrong
94	C2H3N3O	6/6	0.01	[0, 0]	valid-but-wrong
96	C2H3N3O	6/6	0.01	[0, 0]	valid-but-wrong
98	C2H3N3O	6/6	0.01	[0, 0]	valid-but-wrong
104	C2H3N3O	6/6	0.02	[0, 0]	valid-but-wrong
108	C2H3N5	6/6	0.01	[0, 0]	valid-but-wrong
143	C2H4N2O2	6/6	0.02	[0, 0]	valid-but-wrong
203	C2H4N4	6/6	0.02	[0, 0]	valid-but-wrong
258	C2H6O2	6/6	0.35	[1, 0]	ridge-stall
265	C3H2N2O	5/5	0.42	[–, 0]	partial

Eight of the ten reach valid minima on both sides (*valid-but-wrong*): GAD converged to a real TS, IRC descended to two real minima, just not the pair the `Transition1x` dataset labels for that reaction. This is a dataset-side/labeling issue or a case where multiple reaction channels exist from the same saddle.

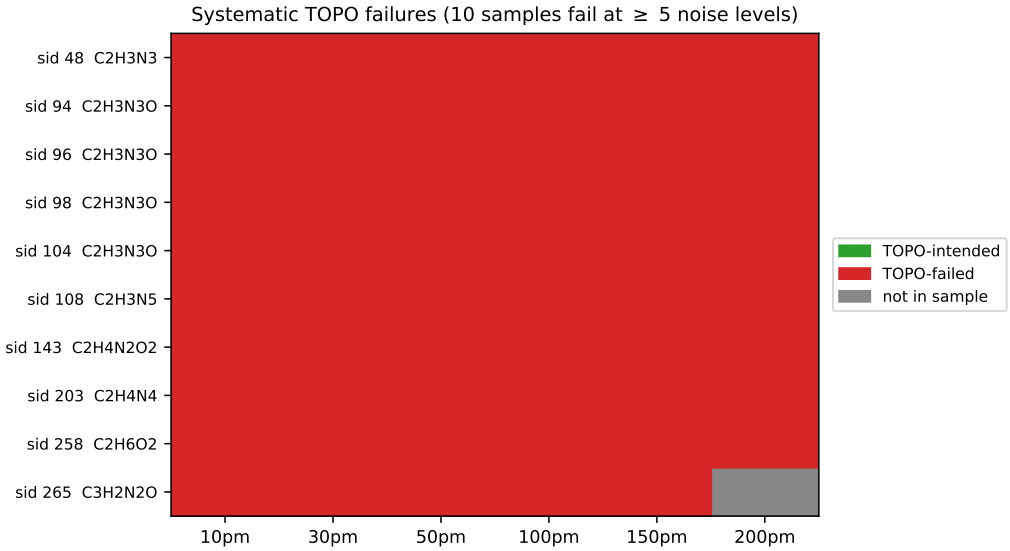

Figure 5: Same-sample TOPO outcome across the 6 noise levels. Persistent red rows flag systematic failures — these are reaction-labeling artifacts, not IRC-integrator stochasticity.

TS Quality vs IRC Outcome

Cross-linking IRC outcome with the upstream GAD convergence trajectory:

Noise	eig_0 median (int)	eig_0 median (fail)	cstep median (int)	cstep median (fail)
10 pm	−3.56	−7.01	95	107
30 pm	−3.46	−7.03	206	220
50 pm	−3.66	−6.65	295	321
100 pm	−4.45	−7.02	454	469
150 pm	−6.48	−2.30	585	886
200 pm	−8.85	−7.64	662	1110

Two observations:

- At low–mid noise, TOPO-failing TSs have *sharper* saddles (more-negative λ_0) than successful ones. Interpretation: these are steeply-curved saddles where IRC escapes quickly into one basin, but the "right" product requires a more gradual descent.
- At high noise (150/200 pm), TOPO-failing TSs converge with substantially *more GAD steps* (`cstep` median +50% above successful). These are ambiguous saddles: GAD struggles to find them, and when it does, IRC often ends up in the wrong basin.

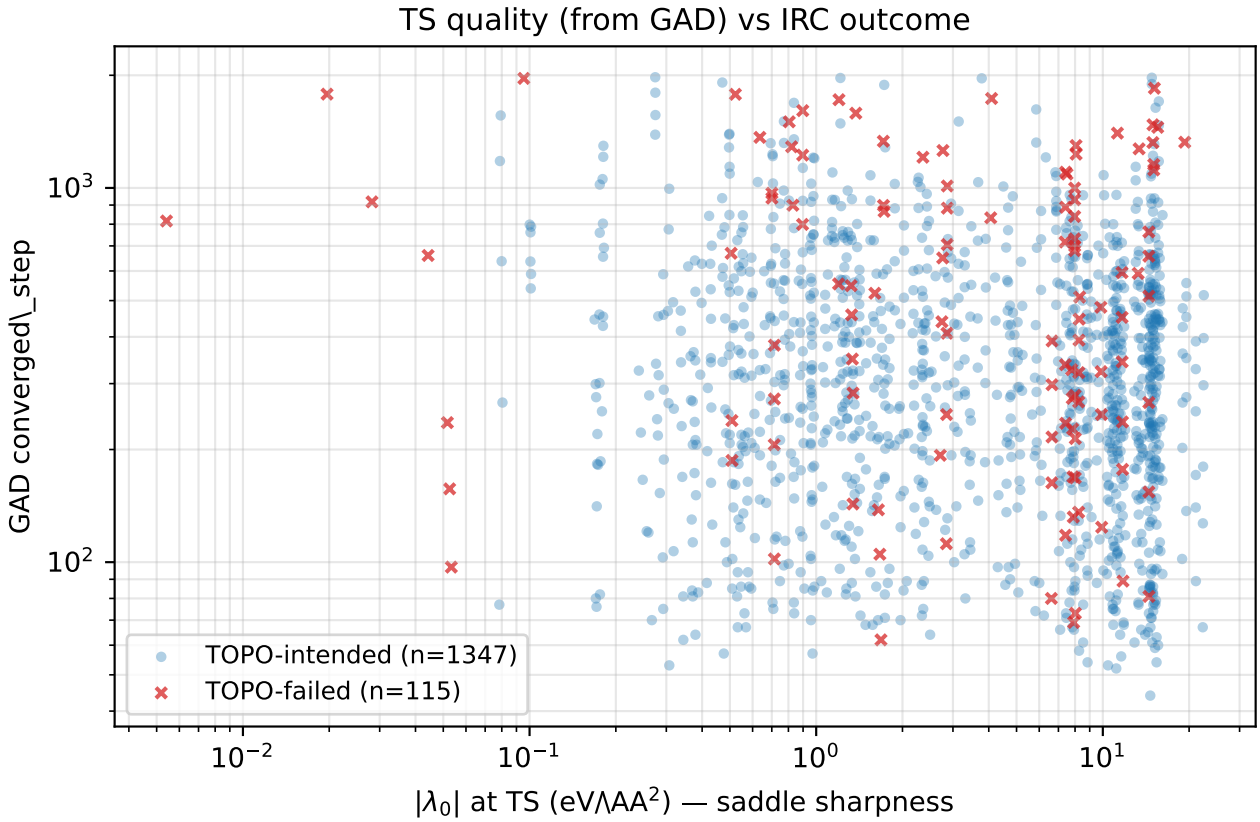

Figure 6: $|\lambda_0|$ at TS convergence vs the GAD step at which it converged. TOPO-failed runs are not concentrated in any single region; they spread across both axes — no simple TS-side quality gate would pre-filter them.

Per-Formula Difficulty

formula	n	TOPO-int%	wall_{avg} (s)
C2N2	4	50.0	19.8
C3H2N2O	15	60.0	15.1
C2H3N5	38	71.1	18.9
C2H6O2	26	76.9	10.6
C2H4N2O	34	82.4	9.3
C2H3N3O	306	88.6	12.3
C2H4N2O2	58	89.7	9.9
C2H5NO	57	91.2	8.2
C2H4N4	263	93.9	8.9
C3H2N2O2	68	94.1	21.0
10 additional formulas are TOPO-intended at 100% ($n \geq 3$).			

Wall Time

Noise	mean	50%	90%	99%	max
10 pm	12.6	9.4	32.1	44.6	49.7
30 pm	12.4	9.5	28.5	44.1	48.6
50 pm	12.3	9.1	28.1	44.0	48.7
100 pm	12.2	9.2	23.1	44.4	49.5
150 pm	11.8	8.1	21.9	44.3	45.5
200 pm	10.4	6.3	17.2	44.2	45.6

Addendum A — Rigorous predictor-corrector IRC on the same TSs

The `rigorous` integrator (Hratchian–Schlegel EulerPC-style predictor-corrector with Eckart projection of gradient and Hessian every step, K=2 hold, adaptive arc-length, initial kick along the lowest *vibrational* eigenvector) was run on the same $n = 1462$ `gad_dt003`-converged TS set as the headline `sella_hip` sweep. Same criteria, same scoring. Wall time ≈116 s/sample (10× slower than `sella_hip`).

Noise	n	TOPO-int%	RMSD-int%	vs. sella_hip TOPO	wall_{avg} (s)
10 pm	284	50.7	34.2	−43.7 pp	115.6
30 pm	283	50.9	34.6	−43.4 pp	116.4
50 pm	276	51.4	35.5	−41.7 pp	117.7
100 pm	262	52.7	37.4	−41.2 pp	116.8
150 pm	214	51.9	40.2	−38.8 pp	117.4
200 pm	143	55.9	46.9	−24.5 pp	115.0
all	1462	**51.9**	**37.2**	−40.2 pp	116.5

Interpretation. Rigorous reaches valid minima in most runs (endpoint-minimum rate is comparable to `sella_hip`), so the underperformance is not integrator instability. It's a *wrong-basin* failure: the

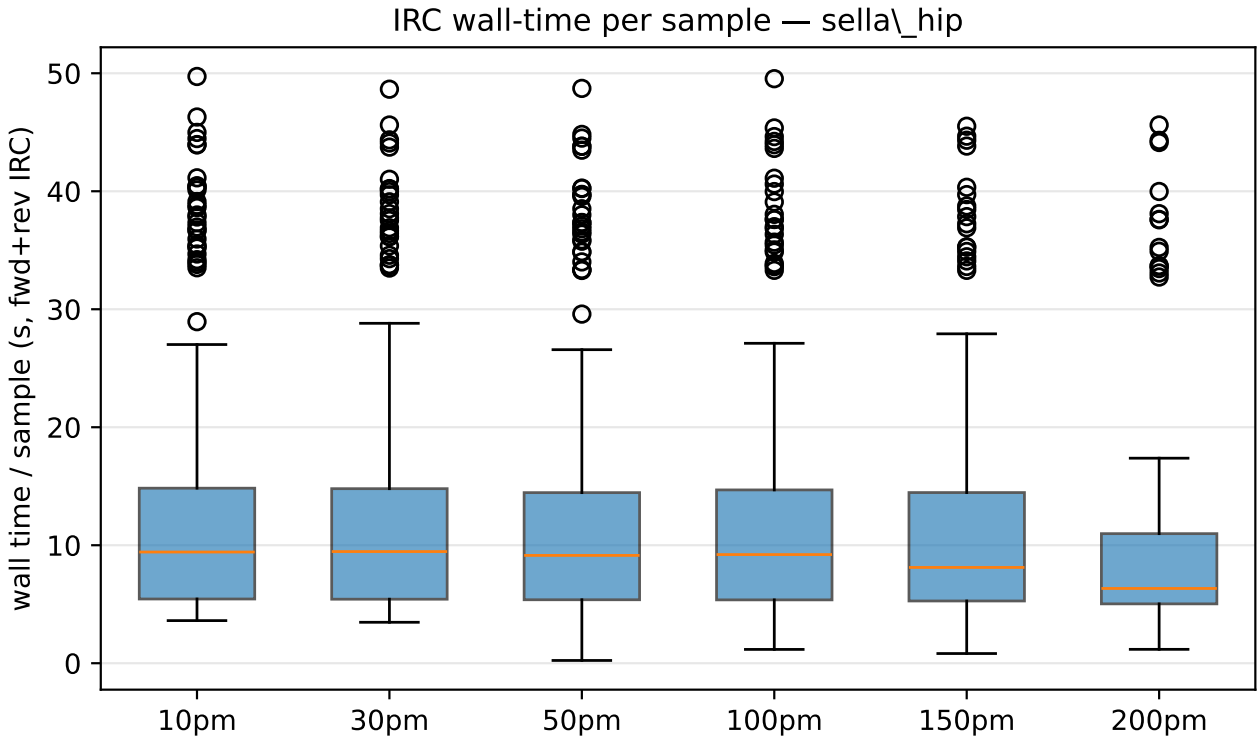

Figure 7: IRC wall time per sample (forward+reverse, up to 500 steps each direction). Typical completion is 5–15 s; the long tail reflects hard cases where Sella's inner loops hit the 500-step ceiling.

predictor-corrector descends to a different real minimum than the one `sella_hip` descends to. The inverted noise dependence (more success at higher noise) suggests the initial-kick direction or the step-size policy is biased near soft saddles where it matters most. Investigation deferred to backburner.

Addendum B — IRC on *all* GAD endpoints (no convergence filter)

The headline sweep restricts to TSs that pass the local gate ($n_{\text{neg}} = 1 \wedge \|\mathbf{F}\| < 0.01$). Dropping that filter and running `sella_hip` IRC on the final-step coordinates of *every* gad_dt003 trajectory (converged or not) tests whether the local gate is actually doing meaningful work and gives a preview of "IRC-as-convergence-criterion".

Noise	n	TOPO-int%	RMSD-int%	TOPO (conv. only)%	ΔTOPO	wall$_{avg}$ (s)
10 pm	300	92.7	62.0	94.4	−1.7 pp	13.7
30 pm	300	92.7	61.3	94.3	−1.6 pp	13.9
50 pm	300	89.7	60.7	93.1	−3.4 pp	14.1
100 pm	300	86.3	58.7	93.9	−7.6 pp	15.5
150 pm	300	69.7	49.3	90.7	−21.0 pp	17.2
200 pm	259	47.9	34.4	80.4	−32.5 pp	18.5
all	1759	79.3	55.0	92.1	−12.8 pp	15.2

Interpretation of the gate's marginal value.

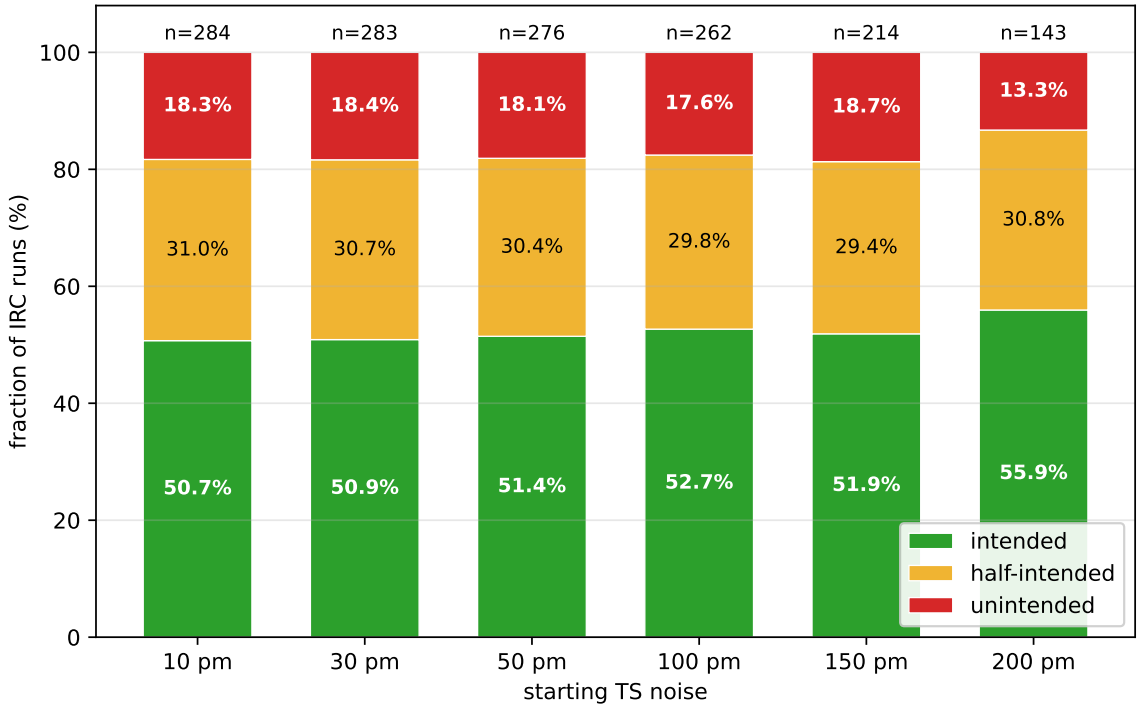

Figure 8: `rigorous` IRC on `gad_dt003` converged TSs — bond-graph topology criterion. Same denominator and same definition of *intended* as the `sella_hip` headline charts. Note the inverted noise dependence: success rate rises slowly with noise, from 50.7% at 10 pm to 55.9% at 200 pm. `sella_hip` does the opposite (94.4%→80.4%).

- At low noise (10–50 pm) the gate barely moves the needle: dropping it loses 1.7–3.4 pp of topology success. GAD converges almost every sample, and even non-converged tails are close enough to a real TS that IRC still finds the right minima.
- The gate's value ramps steeply with noise: at 100 pm it saves 7.6 pp, at 150 pm 21 pp, at 200 pm **32.5 pp**. Samples where GAD failed to converge at high noise are genuinely stuck on bad parts of the PES, and IRC from them doesn't recover.
- This calibrates the proposed "IRC-as-convergence-criterion" direction: at low noise the local gate and an IRC gate will disagree on ∼2% of samples; at high noise they'll disagree on nearly a third.

Addendum C — sella_hip IRC on Sella-optimized TSs

Head-to-head: applying the same sella_hip IRC to TSs found by Sella's own optimizer (`sella_cartesian_eckart_fma` 1000 steps, same noise seeds and same 300-sample universe as gad_dt003) instead of GAD-found TSs.

Implementation note. A naive rerun returned trivial halts — Sella's IRC inherits ASE's "check convergence before first step" pattern, and the Sella-refined TSs already satisfy "fmax < 0.01" (Sella's default IRC halt threshold). A monkey-patch was added to `src/gadplus/search/irc_sella_hip.py::_force_first_` that returns `False` from every convergence check until `step()` has run at least once. The IRC integrator itself is unchanged.

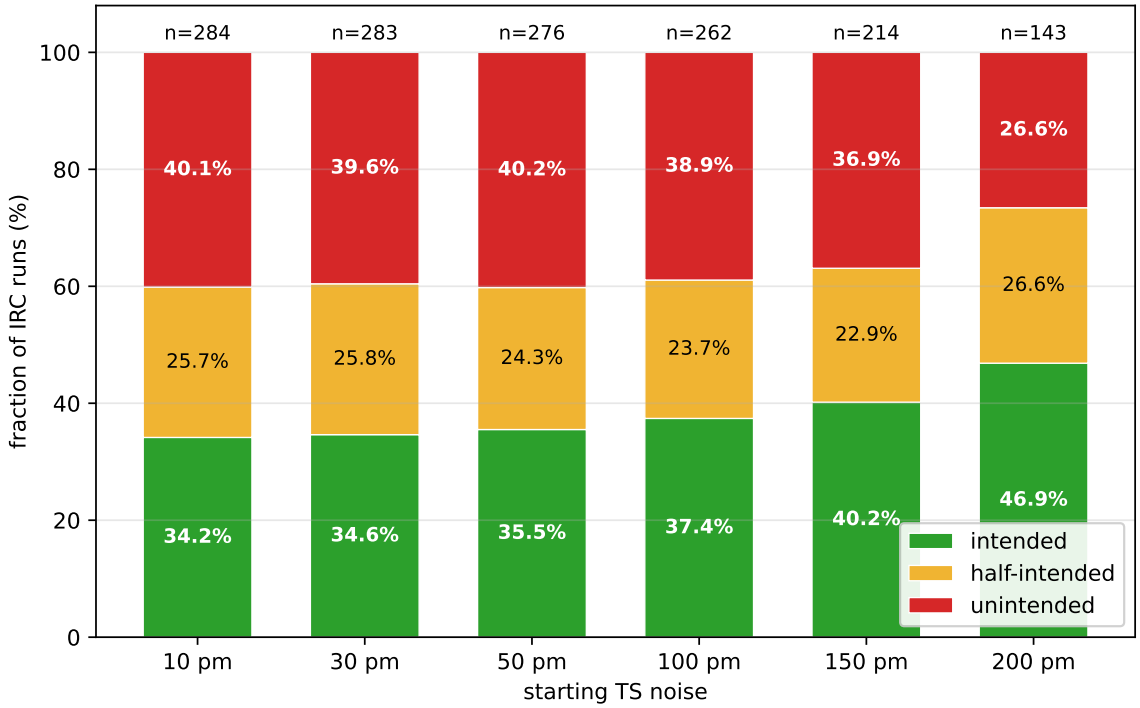

Figure 9: `rigorous` IRC on `gad_dt003` converged TSs — strict RMSD criterion (< 0.3 Å). Same denominator; intended ≡ both endpoints within 0.3 Å RMSD of the labeled (R, P).

Noise	n	TOPO-int%	RMSD-int%	TOPO vs. GAD	n vs. GAD	wall$_{avg}$ (s)
10 pm	284	94.4	64.8	+0.0 pp	±0	12.4
30 pm	283	93.6	64.3	−0.7 pp	±0	12.6
50 pm	274	94.2	65.3	+1.1 pp	−2	12.6
100 pm	236	92.8	63.1	−1.1 pp	−26	12.7
150 pm	159	86.8	59.7	−3.9 pp	−55	12.7
200 pm	83	81.9	61.4	+1.5 pp	−60	11.6
all	1319	**92.2**	63.7	+0.1 pp	−143	12.5

Interpretation.

- **Per-TS IRC-quality is essentially identical.** GAD and Sella both produce TSs that pass the topology criterion at 92% overall with the same sella_hip IRC integrator. Once you restrict to converged TSs, *which* optimizer found them is not the bottleneck — the bottleneck is whether IRC resolves to the labeled (R,P) pair.
- **The difference is how many TSs each optimizer finds.** Sella converges on 284 samples at 10 pm (same as GAD) but only 83 at 200 pm (GAD: 143, i.e. +60 more TSs). Sella's optimizer gives up at high noise where GAD keeps finding saddles. Multiplying rates by sample counts: at 200 pm, Sella delivers $0.819 \times 83 = 68$ topology-valid reactions; GAD delivers $0.804 \times 143 = 115$. GAD is strictly better in absolute yield at high noise.
- **Wall-time parity.** Sella's TS refinement is typically a few steps per sample (1–5 s) vs GAD's ∼100–1000 steps (10–100 s). At low noise, the total (TS + IRC) favors Sella by ∼2×; at high noise, GAD's robustness dominates the tradeoff.

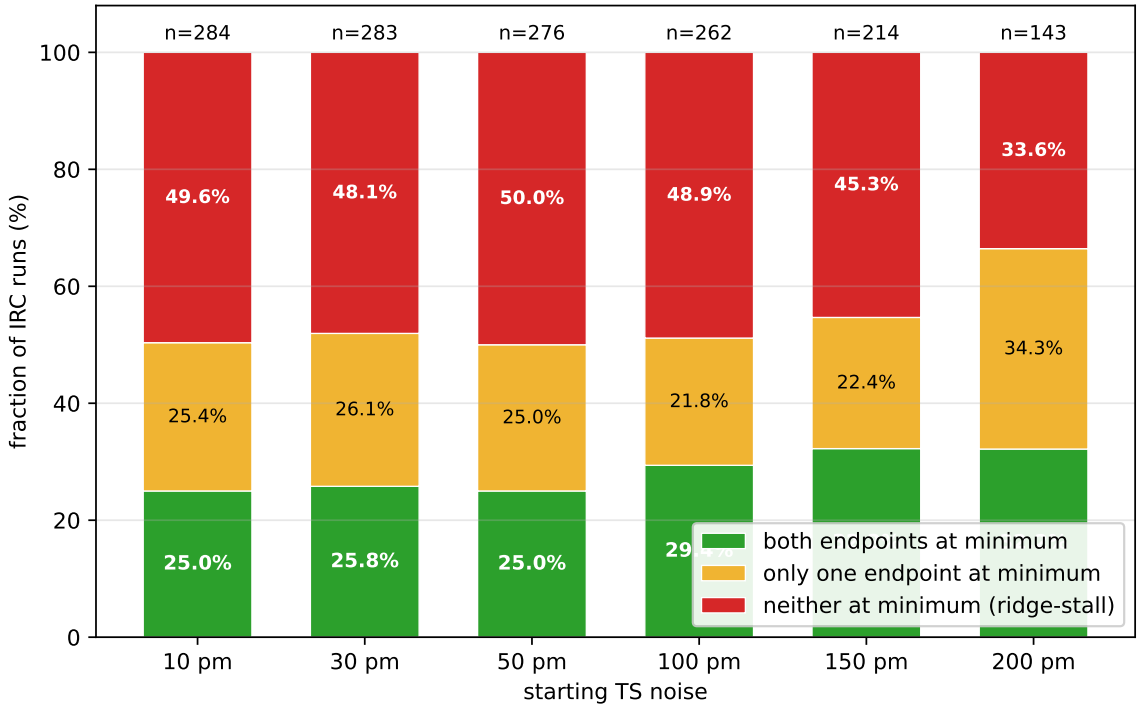

Figure 10: `rigorous` IRC on `gad_dt003` converged TSs — endpoint vibrational quality. Same denominator. The "unintended" bars from the topology chart are *not* ridge-stalls — rigorous reaches real minima at comparable rates to `sella_hip`, just wrong ones.

Summary and Implications

- **92.1% TOPO-intended across all noise levels, with zero errors.** sella_hip is a robust, production-grade IRC on HIP.
- **Strict RMSD-intended is 64.7%.** The ∼27 pp gap is not a noise artefact — it does not improve at lower noise. It is the natural conformational spread of the labeled endpoints.
- **Only 0.5–1.8% of runs end on a ridge** (neither endpoint at a vibrational minimum). Integrator reliability is excellent.
- **5.8% are "valid-but-wrong" minima** — IRC reached real minima, just not the labeled ones. Eight out of the ten persistent-failure samples are this case. This is a dataset labeling / reaction-channel ambiguity, not a method failure.
- **Degradation only kicks in at 200 pm**, and even there TOPO-intended is >80%.
- Failing samples are spread across $|\lambda_0|$ and cstep at the TS side — there is no single easy filter that would predict IRC-failure from GAD-side telemetry.
- **Wall time is modest:** median ∼9 s per sample across fwd+rev. The full 1462-run sweep finished in ∼30 min wall parallelized over 6 MIG slices.

Next. The rigorous predictor-corrector IRC sweep (Hratchian-Schlegel EulerPC-style) is running in parallel (job 59464202, ETA ∼11 h). Direct head-to-head against sella_hip on the same $n = 1462$ TS set.

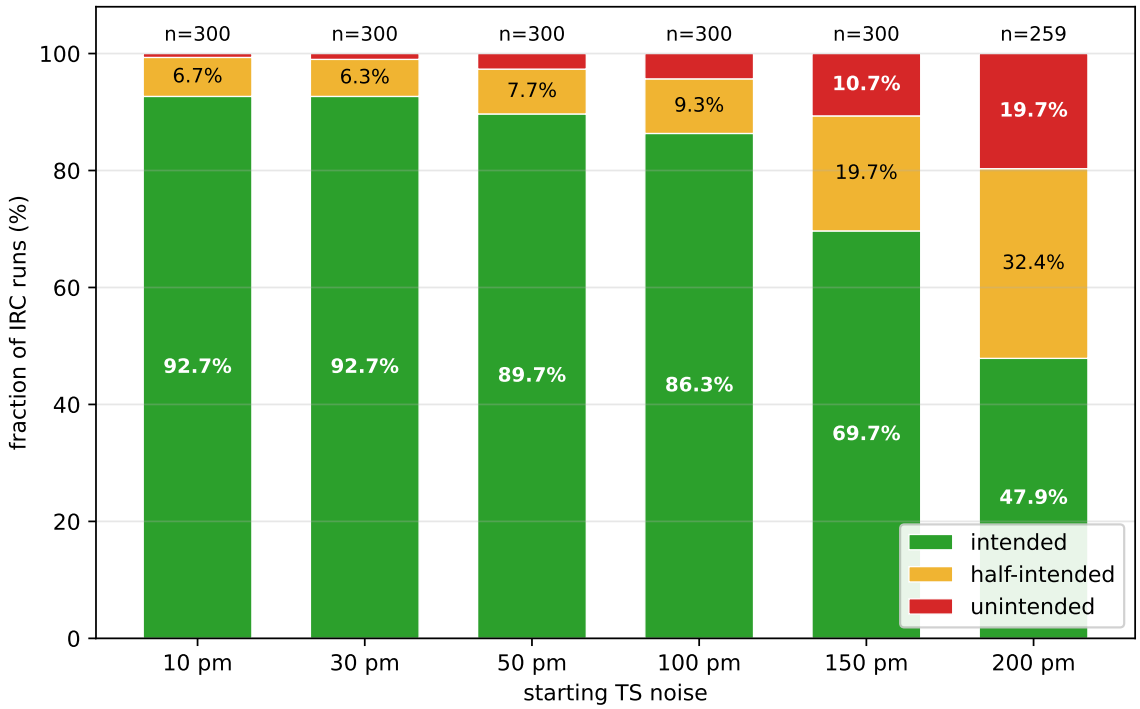

Figure 11: `sella_hip` **IRC on *all*** `gad_dt003` **sample endpoints — bond-graph topology criterion.** Denominator = full 300 samples per noise level (259 at 200 pm, since only 259 GAD trajectories existed there). No local-gate filter applied. Same intended/half/unintended definition as the headline charts.

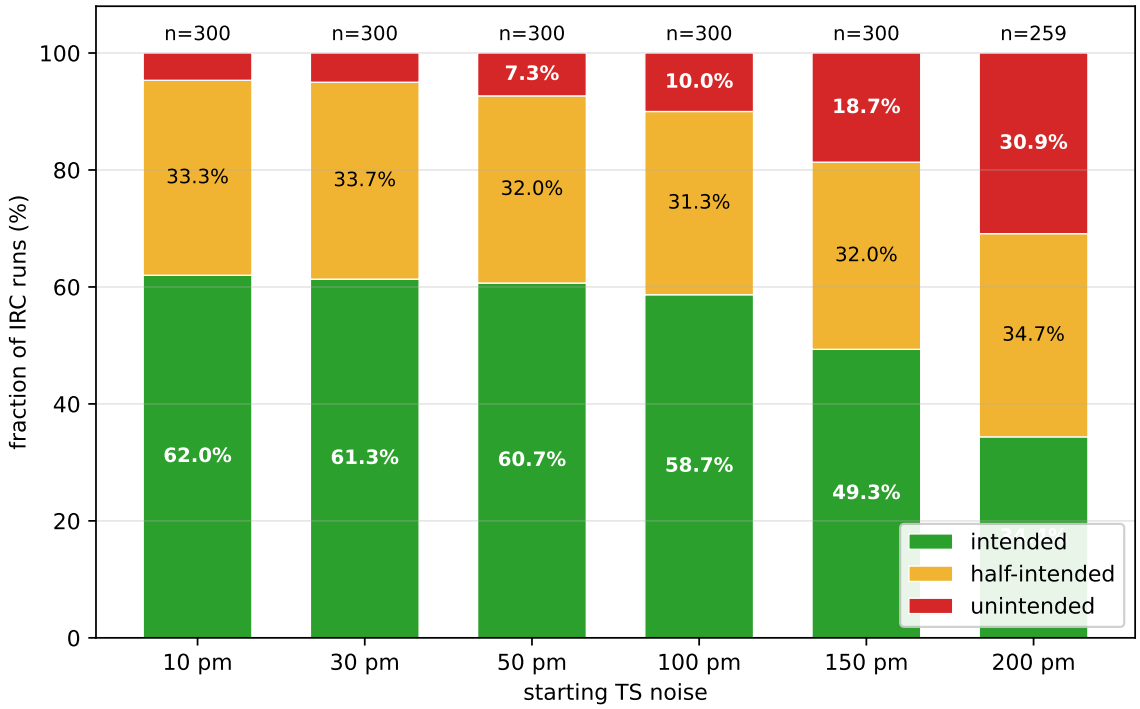

Figure 12: `sella_hip` **IRC on *all*** `gad_dt003` **sample endpoints — strict RMSD criterion (< 0.3 Å).** Same denominator; same definition of intended as the headline RMSD chart.

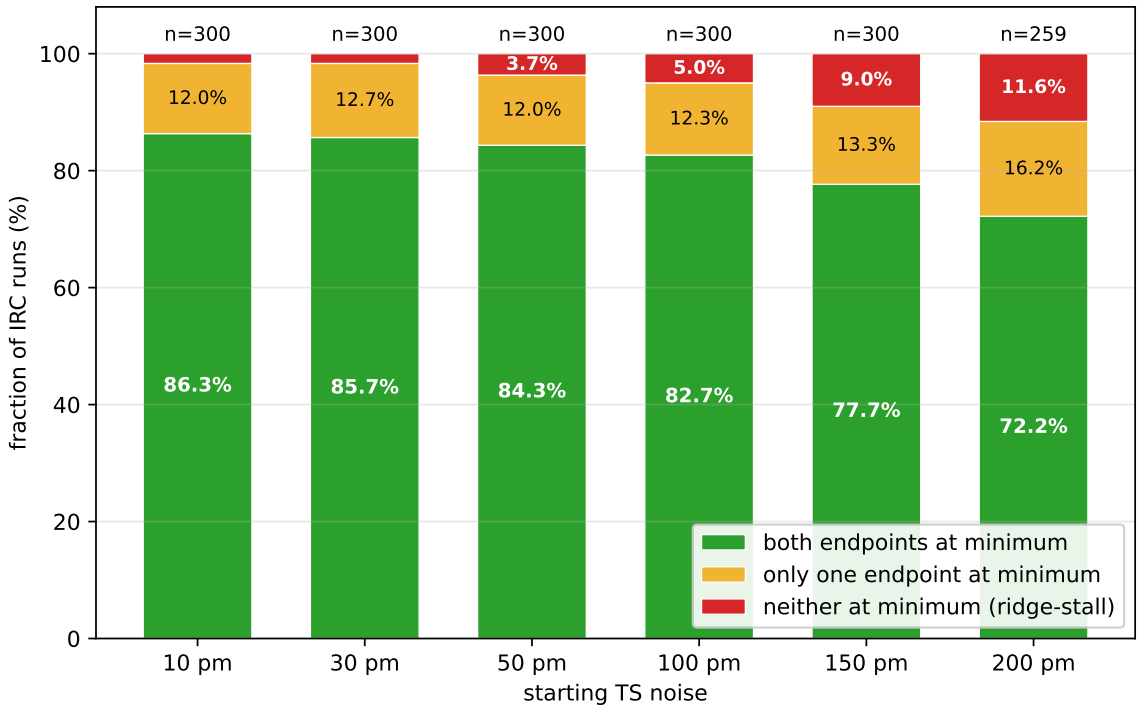

Figure 13: `sella_hip` IRC on *all* `gad_dt003` sample endpoints — endpoint vibrational quality. Same denominator; both / one / neither endpoint at a true minimum ($n_{\mathrm{neg,vib}} = 0$).

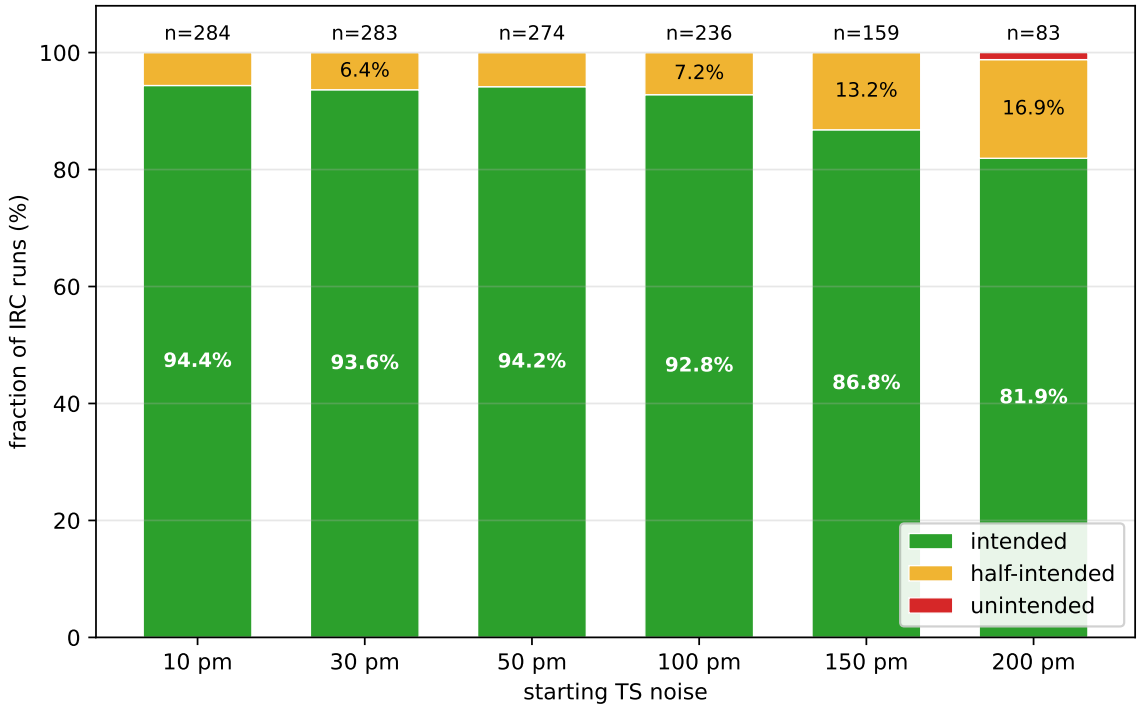

Figure 14: `sella_hip` IRC on Sella-optimized TSs — bond-graph topology criterion. Denominator = `sella_cartesian_eckart_fmax0p01` converged TSs at each noise level. Same definition of *intended* as the headline chart (direction-agnostic bond-graph match on both sides).

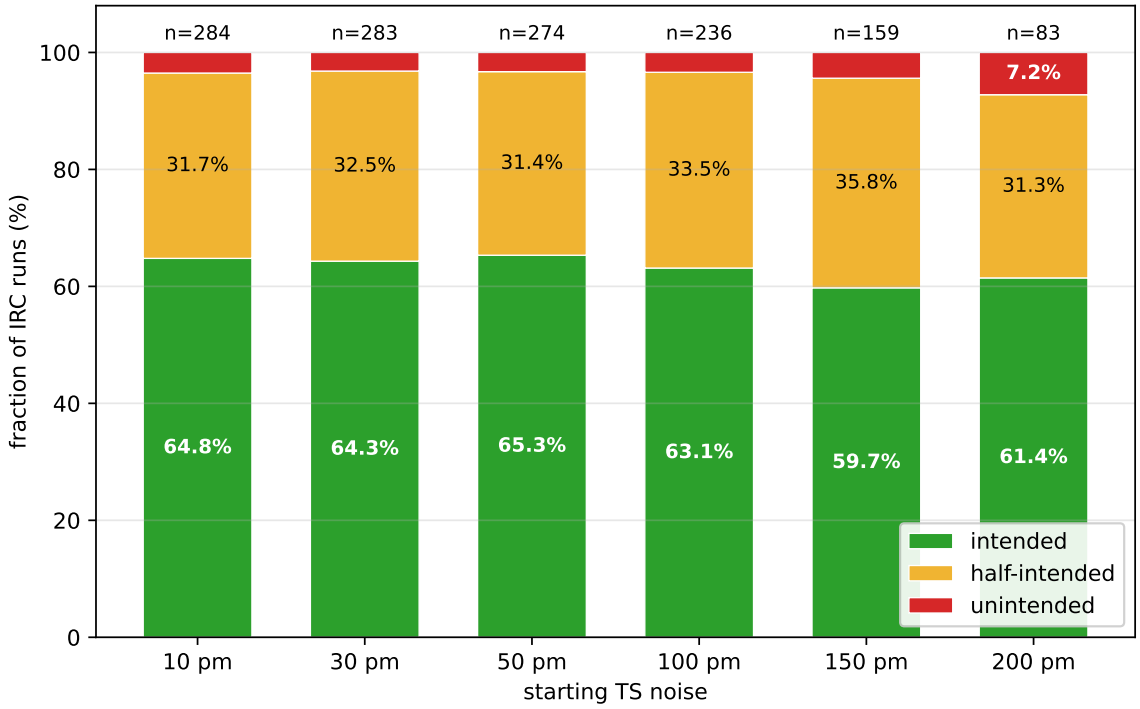

Figure 15: `sella_hip` IRC on Sella-optimized TSs — strict RMSD criterion (< 0.3 Å).

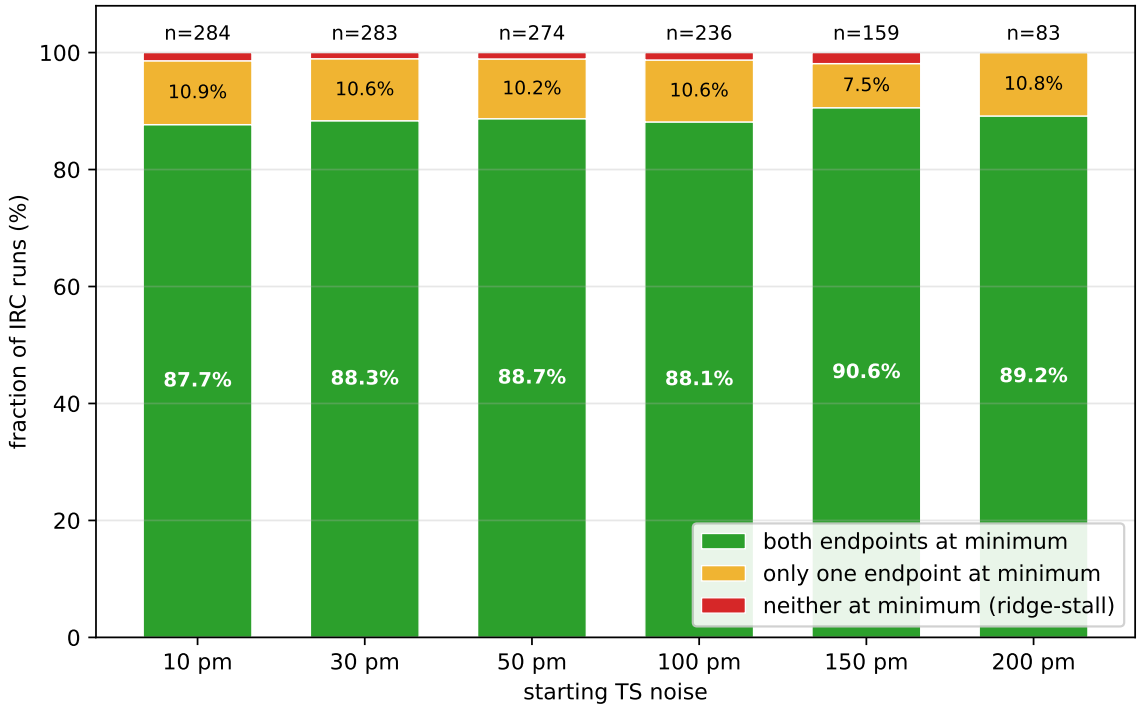

Figure 16: `sella_hip` IRC on Sella-optimized TSs — endpoint vibrational quality.